

-
-
-
-
-

Chapter 4

Drinks EDI Application Interface

Overview



Introduction

The use of Electronic Data Interchange (EDI) can bring your organisation many measurable benefits:

Speed

EDI sends and receives documents quickly. There is none of the delay associated when mailing documents.

Accuracy

Since EDI transmits documents electronically, there is no need to re-enter the information, thus reducing keying errors.

Cost Savings

The quick turn-around of documents reduces inventory overhead and increases the available cash flow. EDI also reduces mailing costs.

Productivity

Employees are no longer needed to enter documents or address, stamp and mail envelopes, leaving them free to perform other tasks.

The System 21 Drinks EDI solution has two closely integrated components. These are:

- The System 21 Drinks EDI module
- Premenos EDI enabler

Together, these products satisfy the requirements of a successful EDI installation, namely:

- Controlled Application Interfaces
- Standard Message Format Support
- Conversion Facilities
- Communications Management Facilities

EDI provides support for the most widely-used Message Standards and Value Added Networks. It will also give mailbox access and document conversion facilities. The Drinks EDI module supplies the facility to import and export documents - in a controlled fashion - to and from the standard Sales Order Processing and Purchase Management modules.

Overview

You can use System 21 EDI to import and export of documents to and from Sales Order Processing and Purchase Management. These documents, traditionally sent and received in paper form, are exchanged as electronic messages.

The application will support:

- Inbound Sales Orders
- Outbound Acknowledgements
- Outbound Pick Notes to Third Party Trading Partners
- Outbound Advice of Despatch
- Outbound Sales Invoices
- Outbound Purchase Orders
- Outbound Sales Catalogues
- Outbound Product Catalogues
- Inbound Purchase Order Acknowledgements
- Inbound Advice of Despatch from Suppliers and Third Party Trading Partners
- Inbound Third Party Stock Balances
- Inbound Third Party Stock Adjustments
- Inbound Product Activity

The application can be used in two ways.

The application is designed to work very closely with the Premenos EDI enabler, EDI/400. This caters for the conversion of messages into and out of standard message formats. The Drinks EDI module adds new function for users of Sales Order Processing and Purchase Management in the shape of direct calls to EDI. So, for example, one interface option will allow the preparation of electronic purchase orders, and will also invoke a routine in EDI to convert those orders into standard message formats for onward transmission to suppliers. This close-coupled link to EDI removes the need for repeated manual intervention in the conversion and transmission process.

The second potential use for the interface module is as an electronic batch transaction facility. For example, it is common in certain industries for sales orders to be captured by the field sales force on portable microcomputers. Traditionally, these have then been re-entered onto the central Sales Order Processing application. With System 21 EDI, this is no longer necessary. Using existing facilities, the orders can be loaded up onto the host system, and taken into the order bank, in a controlled fashion, using the Drinks EDI module.

This module applies the same validation logic to inbound and outbound messages as is used in the conventional operation of the system. Full exception and audit reporting is provided.

Premenos EDI

Premenos EDI provides the following functions:

- Reformatting of messages between the internal format of System 21 and the formats of the major international EDI Standards.
- Communication modules to enable direct and straightforward connection to all the principal Value Added Networks.
- Support for point-to-point communications where a Value Added Network is not in use.
- Full transaction logging, error recovery and security features.

Support Of EDI Standards

EDI will support the following national and international EDI Standards:

- ANSI X12
- INCOME
- EDIFACT
- ODETTE
- Tradacoms
- TDB
- US

In addition, Premenos EDI contains a Standard Modification Tool, which will allow modification of Standards already defined to the product, and definition of new Standards.

Value Added Network Support

Premenos EDI is supplied with a module enabling connection to the Premenos Information Network. In addition, further communication modules are available to enable connection to any of the following Value Added Networks:

- INS/TRIDENT
- AT&T Easylink
- GEIS network services
- McDonnell Douglas network services
- Control Data/Redinet
- Sterling/Ordernet Version 3
- Kleinschmidt network services
- TranSettlement/Transend
- General Motors/GMNET

As well as the Value Added Networks (VAN) listed above, EDI will allow direct point-to-point communications between systems, where a VAN is not used.

Relationship To Other Applications

The following applications are pre-requisite to the Drinks EDI module:

- Drinks Inventory Management
- Drinks Sales Order Processing (for Sales documents only)
- Drinks Purchase Management (for Purchasing documents only)
- Application Manager

The application is designed to be able to add translated documents into the Mailbox Send Queue in the EDI application automatically. This requires, however, the trading partner being defined to both Drinks EDI and Premenos EDI.

Application Configuration

To set up the additional data required for sending and/or receiving documents via EDI, you need to create and maintain certain elements.

The main elements created and maintained within Drinks EDI are:

- Documents
- Trading Partners
- Location References

The main elements created and maintained within Premenos EDI are:

- Translation Maps
- Code Translation Tables

This documentation describes the Drinks EDI component. For more information about the set up and use of Premenos EDI, see the Premenos User Manual.

Documents

All document types to be sent via EDI must be defined in the Application Interface, for example, Purchase Order, Sales Invoice. Within this, the document ID must be specified for each EDI message standard used.

These definitions will be shared by all companies defined to the application.

Trading Partners

All trading partners - customers, suppliers or third parties, who are to be communicated with via EDI must be defined in the Application Interface.

Within this, the document types to be sent to each partner must be selected from those defined above. At this point you specify if the document is to be printed and/or sent via EDI for this trading partner.

Further details must be defined for the Trading Partner in the EDI application. These are required to support the document translation and communication functions in EDI.

Location References

Where delivery is required to multiple locations, a set of agreed location reference codes must be maintained in order to establish the correct delivery address.

Translation Maps

A translation map is required for each trading partner/document combination, whether they are sent or received. This is used by EDI to map data sent via EDI from the Application Interface database into the appropriate EDI message standard, and vice-versa for data received.

Code Translation Tables

A code translation table is required to convert codes, e.g., Units of Measure, Sales Tax, into the equivalent codes recognised by the appropriate message standard when sending documents.

Conversely, they are used to translate EDI standard codes into codes for receiving documents.

For more details on setting up trading partners see Trading Partners Maintenance and Third Party Trading Partners Maintenance.

Processing Cycle

The following example is provided to illustrate how the EDI processing cycle is supported by the Premenos EDI enabler. The example given is that of an incoming sales order. The procedure for outgoing messages is similar.

Although each step can be executed separately, the products can be configured to allow steps 1, 2 and 3 to be combined into one option leaving step 4 to be executed separately.

Step 1 Access Value Added Network

Having agreed with the applicable electronic trading partners which VAN is to be used to exchange messages, you can define it to Premenos EDI. This definition tells Premenos EDI how the connection to the VAN should be established.

Premenos EDI supports binary synchronous communications with all the networks for which communications modules are provided. The selected protocols can be used over leased or dialled lines. In the case of dialled connections, either autodial or manual dial is supported. During set-up, other parameters concerning communications with the network will be recorded. Premenos EDI stores these parameters with the telephone numbers for dialled connections.

For this example it is assumed that EDI has established the to the AS/400.

Step 2 Transfer Documents

Once the communications session is started, Premenos EDI will use the previously defined parameters to control the conversation with the network. Premenos EDI will then collect any messages in the network mailbox and transfer them down to its own mailbox on the AS/400. It also checks if there is any outbound mail, for example, acknowledgements that should be sent, and if there are it will transfer them up to the network mailbox. Premenos EDI will then distribute these to the appropriate trading partners. Premenos EDI maintains an on-line log containing the details of each communications session.

Once the transfer process is complete, Premenos EDI will - if appropriate - disconnect from the network.

The received message will now be sitting in the EDI mailbox. It can be displayed, printed or deleted while it is there. It can also be translated and moved into the Drinks EDI module.

Step 3 Translate To In house Format And Files

Before Premenos EDI can successfully translate an incoming order out of its EDI format and into the in-house Drinks EDI format, a number of definitions must have been set up within Premenos EDI:

- **Trading Partner Transaction Type**

This defines a unique business relationship regarding the two trading partners concerned in the transaction of a particular document, in this case an order. Premenos EDI uses this information when handling received messages. The information is also used when addressing and sending outbound messages.

Premenos EDI also uses the transaction type to identify the translation map that

should be used to translate the document (whether inbound or outbound). The transaction type records other information regarding the relationship that has been agreed with a trading partner, such as the network over which the document should be exchanged, the trading partner's network ID, and version and release values for the standard used.

- **Application Interface File Definitions**

The Drinks EDI module files that will receive the incoming order must be defined to Premenos EDI as 'user files'. During translation, Premenos EDI directs the translated data to these files.

- **Translation Map**

The relationship between the incoming Tradacoms order and the Drinks EDI module files has to be defined. To do this, a 'map' must be created. This map lies at the heart of the translation process. Premenos EDI moves data contained in the Tradacoms standard order into the corresponding Drinks EDI module files. A map is required for all EDI documents that are to be sent or received. This is true of all EDI standards.

- **Code Translation Tables**

EDI standards will often have pre-defined codes for certain data elements. These must be translated so as to be accepted as valid. Similarly, if your trading partner uses different values (e.g. UOM, delivery address or product codes), they must also be translated into values acceptable to the particular application. Hence code translation tables. Premenos EDI uses these tables to translate standard and trading partner-specific codes into values that are acceptable.

The translation process takes all these elements into account. When completed, the translated order will be written to the appropriate Drinks EDI file.

Step 4 Application Interface

Once Premenos EDI has successfully translated the incoming order and placed it in the Drinks EDI module file; a Drinks EDI option can be executed to place it on the Sales Order Processing order bank. This interface program performs the same validation upon the incoming orders that is applied normally to orders entered in to the system via the keyboard. Exceptions will cause the order to be suspended for further investigation. In the case of major exceptions, the order will be rejected. Full exception and control reports are provided.

The logic described in this example is applied to all inbound or outbound documents supported by Premenos EDI and the Drinks EDI module. In the case of outbound documents, for example purchase orders, most of the validation will already have been carried out by the application generating the document. For outgoing documents, Premenos EDI and the Drinks EDI module examine the validity of the document as an EDI message, deciding whether they should be sent via EDI and/or printed.

Third Party Trading Partners

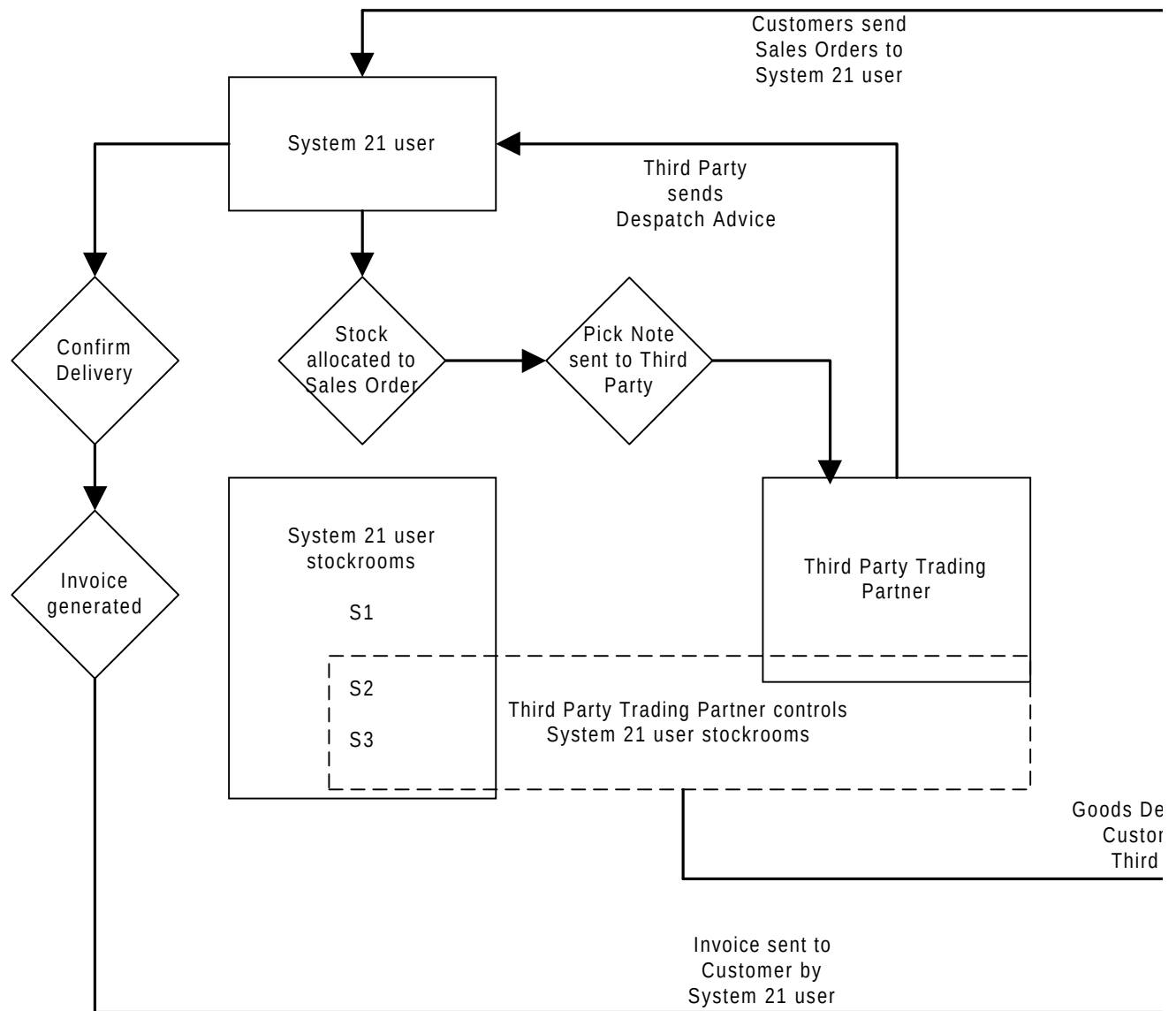
Drinks EDI includes facilities to transmit and receive EDI documents from Third party trading partners. Third party trading partners are organisations which manage stockrooms for the organisation using System 21.

The System 21 user takes orders from customers and sends picking notes to the third party trading partner. The third party trading partner manages the picking and despatch of the goods to the customer and sends despatch advises back to the System 21 user.

Similarly, the System 21 user places stock purchase orders with suppliers who deliver the stock to the third party controlled stockrooms. The third party trading partner then sends goods received notes back to the System 21 user.

The following diagrams illustrate these scenarios:

EDI and Customers



EDI and Suppliers

